

Combined Enhanced Conformational Sampling and Protein-Ligand Dockings on CYP3A4 Improves in Silico Prediction of Drug Metabolism

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Abstract

Cytochrome P450 3A4 (CYP3A4) is the primary drug-metabolizing enzyme in humans. This is largely due to its highly flexible structure, which enables it to bind a broad spectrum of xenobiotics with a wide variety of shapes and physicochemical properties. This plasticity, however, presents challenges for in silico substrate prediction methods, in particular those that consider only a limited conformational space of the receptor or that rely on static crystal structures. In this study, we aim to explore the conformational landscape of CYP3A4 using Gaussian accelerated Molecular Dynamics (GaMD) and then to assess the viability of using this method to identify pro-reactive conformations through protein-ligand docking. GaMD simulations were initiated from distinct X-ray structures in triplicate 500 ns runs. The results demonstrate that the simulations cover a very large conformational space, including all reported X-ray structures as well as conformations with significantly expanded active-site geometries unseen in the experimental data. Protein-ligand docking of known CYP3A4 substrates onto the main cluster representatives from GaMD simulations allows the identification of metabolically active poses of the substrates. This is true even in cases where substrate-bound X-ray structures or docking against a single structure fails to yield productive results, as in the case of CYP3A4–Erythromycin. Furthermore, a classifier trained on ensemble docking scores discriminates CYP3A4 substrates from non-substrates better than any single conformation (ROC-AUC 0.735 vs 0.697), with performance increasing with ensemble size. Overall, the study shows that enhanced conformational sampling is crucial for accurately modelling highly flexible enzymes like CYP3A4 and can significantly improve metabolic predictions.

Keywords

Cytochrome P450 3A4, Gaussian accelerated molecular dynamics, Ensemble Docking, Conformational sampling, Drug metabolism

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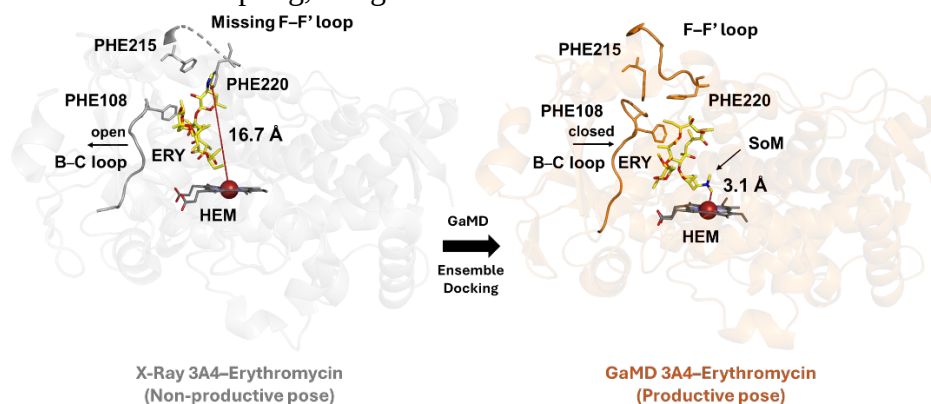
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Overall, the study shows that enhanced conformational sampling is crucial for accurately modelling highly flexible enzymes like CYP3A4 and can significantly improve metabolic predictions.

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1. Introduction

Cytochrome P450 (CYP) enzymes constitute a large, phylogenetically widespread superfamily of heme monooxygenases that mediate the oxidative metabolism of endogenous compounds and xenobiotics [1], [2]. Among human isoforms, CYP3A4 is particularly important because it is responsible for the biotransformation of approximately 50% of clinically used drugs and can accommodate a chemically diverse range of substrates [3], [4], [5]. This broad specificity arises from a malleable active site and pronounced conformational plasticity in key structural elements that govern substrate access, orientation, and product selectivity [6], [7], [8].

Although more than 120 human CYP3A4 crystal structures have been solved, it is not clear that the full ensemble of conformational space has yet been characterized. Conformations can remain very close to each other while accommodating substantially different ligands [9] ; some structures present global conformational changes suggesting that ligand binding occurs via a conformational selection processes [10], and other structures present what appears to be induced fit [11] or even hold multiple ligands in their binding pocket [12]. In some experimental structures, the ligand binds far from the catalytic site, at the entrance of the supposed binding pathway [13]. This wide variety of conformational states leads to large uncertainties for structure-based docking, since calculations are often performed using a single static protein conformation, implicitly assuming that one structure can represent the catalytically competent binding site. For CYP3A4, this assumption is often insufficient: ligands can stabilize distinct induced-fit states, and productive binding geometries may be transient or absent from available experimental structures [14], [15]. As a result, docking against an isolated structure may miss metabolically relevant poses, particularly for bulky or flexible substrates, motivating the development of approaches that explicitly account for conformational heterogeneity through ensemble representations [16], [17].

Prior molecular dynamics (MD) studies have characterized CYP3A4 flexibility, particularly the dynamics of the F/G and B'C loops that gate the substrate access channel [18], [19], [20]. Enhanced-sampling approaches have also been applied to CYP3A4, including metadynamics and Gaussian accelerated molecular dynamics (GaMD) to probe ligand-induced conformational changes [21], [22], [23]. Ensemble docking approaches have been explored for flexible enzymes, including CYP3A4 (ref) but systematic application of enhanced-sampling-derived ensembles to structure-based substrate prediction and site-of-metabolism geometry assessment remains limited.

Enhanced-sampling MD offers a practical route to generate such ensembles at atomistic resolution. Accelerated MD (aMD) is particularly well-suited because it requires no predefined collective variables and efficiently samples rare transitions on accessible timescales [24], [25]. Among these methods, GaMD improves on conventional aMD by applying a boost potential that follows an approximately Gaussian distribution, enabling more accurate free-energy reweighting via cumulant expansion. Coupling GaMD-derived representative conformations with protein–ligand docking then enables direct assessment of whether sampled conformations support catalytically productive substrate positioning relative to the heme iron.

Here, we map the conformational landscape of CYP3A4 by running GaMD simulations initiated from distinct experimentally determined starting points and extracting an ensemble of 13 cluster representatives spanning induced, apo, and open pocket states. We then evaluate this ensemble through ensemble docking of eight known CYP3A4 substrates with established sites of

metabolism, assessing whether they adopt catalytically relevant geometries. This workflow captures the diversity of experimentally observed conformations while extending beyond them to reveal additional active site arrangements and enables identification of productive binding poses in challenging cases where docking to a single X-ray structure is insufficient. More broadly, this work provides an ensemble-aware structural foundation that can be integrated into subsequent metabolism-prediction pipelines, for example by combining ensemble-docking descriptors with modern AI/ML models to improve site-of-metabolism and substrate-classification performance.

2. Computational Methods

2.1. Analysis of experimental CYP3A4 structures

All experimentally determined structures of human CYP3A4 were retrieved from the RCSB Protein Data Bank (<https://www.rcsb.org>; accessed 15 November 2025) using UniProt accession P08684, yielding 121 entries [26]. Each structure was pre-processed with PyMOL v3.1.0 [27] by removing crystallographic waters and retaining only chain A; the heme cofactor and co-crystallized ligands were preserved to maintain native active site geometry. Pairwise C α root mean square deviation (RMSD) values were computed across the curated set following least-squares superposition onto conserved secondary structure elements (helices A–L and the β -sheet core). The resulting 121 $\times$ 121 distance matrix was used to characterize the conformational diversity of the experimental ensemble and to guide the selection of starting structures for GaMD simulations.

2.2. Gaussian accelerated molecular dynamics

A bonded model of the ferric heme cofactor and its axial Cys415 ligand was built with the MCPB.py utility [28] implemented in AmberTools24 [29]. Atomic point charges were assigned by RESP fitting [30], and bonded parameters (bond, angle, and dihedral force constants together with equilibrium values) were extracted from the Hessian of the optimized metal-center cluster using the Seminario approach [31]. Geometry optimization and harmonic frequency calculations were performed in Gaussian09 [32] at the B3LYP-D3 [33], [34] level of theory; the iron atom was described with the Stuttgart–Dresden effective core potential [35] and the remaining atoms with the 6-31G(d,p) basis set. Co-crystallized ligands were parameterized with antechamber [36] using AM1-BCC [37] charges under GAFF [38], and force-field parameters were extracted with parmchk2 [29]. Systems were assembled in tLEaP [29], immersed in a cubic TIP3P [39] box with a 10 Å solute–wall buffer, and neutralized with Na⁺/Cl[−] counter-ions. Each system was minimized, then progressed through the layered equilibration protocol implemented in GaMD-OpenMM: 1 ns of conventional MD preparation, 10 ns of conventional MD for collecting potential-energy statistics, 2 ns of GaMD equilibration preparation, and 50 ns of GaMD equilibration with iterative refinement of the boost parameters.

GaMD applies harmonic boost potentials that smooth energy barriers and accelerate conformational transitions without requiring predefined collective variables [23]. Dual-boost GaMD [40], with independent upper-bound boost potentials on the total potential and dihedral energy terms ($\sigma_0 = 6.0$ kcal mol^{−1} for both), was carried out with GaMD-OpenMM [41] (<https://github.com/MiaoLab20/gamd-openmm>) running on OpenMM [42], using the AMBER topology and coordinate files generated by tLEaP as direct input. Production runs employed a Langevin integrator ($\gamma = 1.0$ ps^{−1}) with a 2 fs timestep, particle mesh Ewald electrostatics with a 10 Å real-space cutoff, bonds to hydrogen constrained, and a Monte Carlo barostat ($P = 1$ bar,

frequency 25 steps) at 300 K. Three independent 500 ns production runs were generated from each of three CYP3A4 structures selected to span the experimental conformational landscape: 5TE8 [43] (midazolam-bound, induced-fit), 5VC0 [44] (ritonavir-bound, open), and 9BV5 [45] (apo-like). This yielded nine trajectories and 4.5 μ s of aggregate sampling.

2.3. Ensemble construction

A two-stage clustering protocol was used to compress the 4.5 μ s of GaMD sampling into a tractable ensemble for docking. In the first stage, each sub-system (induced-fit, apo, and open) was independently clustered with cpptraj [46] on the merged three-replicate trajectory. Solvent and counter-ions were stripped, coordinates were retained every 50 ps, and C α RMSD on the protein core (residues 1–396) was used as the distance metric. Hierarchical agglomerative clustering with average linkage [47] was performed with the number of clusters set to 25. Cluster representatives were extracted as the cpptraj-defined cluster medoid.

In the second stage, the 75 sub-system representatives were aligned onto a common reference (the apo-system medoid) in PyMOL [27] using C α atoms, after which a global all-versus-all C α RMSD matrix was computed over the 463 common-core residues with MDAnalysis [48]. The matrix was clustered hierarchically using Ward linkage [49] as implemented in *scipy.cluster.hierarchy* [50], with the dendrogram cut at 2.25 Å RMSD, yielding 13 final cluster representatives partitioned across the three state types: induced-fit (C1–C7), apo (C8–C9), and open (C10–C13). The cluster medoid (the snapshot with minimum mean RMSD to its cluster mates) was retained as the representative structure. The 2.25 Å cutoff was selected from a coverage scan over cutoffs of 2.0–4.0 Å (Table S1) as the smallest value yielding 100 % coverage of all three sub-system trajectories within the cutoff radius.

2.4. Ensemble docking

Receptor preparation. Each of the 13 cluster representatives was prepared for docking by removing crystallographic waters, retaining the heme cofactor, adding polar hydrogens, and converting to PDBQT format with Meeko v0.7.1 [51].

Ligand dataset. Test compounds were drawn from the curated CYP3A4 reference set of Ni *et al.*, 2025 [52] which compiles 1,343 experimentally verified 3A4 substrates and 1,622 non-substrates. Compounds were imported as SMILES, expanded to three-dimensional conformers with Open Babel v3.1.0 [53], and energy-minimized under the MMFF94 force field [54]. Partial charges and AutoDock atom types were then assigned with Meeko v0.7.1 [51], producing the PDBQT files used for docking.

Docking. Docking was performed with AutoDock Vina v1.2.6 [55], using a rectangular search box of 24.3 $\times$ 24.2 $\times$ 17.8 Å centered on the heme iron of the aligned cluster ensemble (x = 28.93, y = 52.98, z = 45.06 Å), with the default exhaustiveness of 8 and the five top-ranked binding modes retained per substrate–conformation pair.

Classification. Each ligand was represented by its best-pose affinity in each of the 13 conformations. Ligands for which docking returned no valid pose (indicating a failed run) in one or more conformations were excluded, giving 2,617 ligands (1,223 substrates); complete-case analysis was used because these failures were class-dependent, so imputation would introduce label-correlated signal. A logistic regression classifier (scikit-learn v1.9.0) was trained with default parameters, without feature scaling or hyperparameter search. Catalytic geometry was described by the heme iron–to–site-of-metabolism distance, assigned by chemistry rules applied identically to both classes and averaged across the 13 conformations.

Statistical analysis. Performance was assessed by stratified 5-fold cross-validation with ROC-AUC on pooled out-of-fold predictions, confidence intervals from 5,000 bootstrap resamples, and comparisons between correlated ROC curves by DeLong test. Ensemble-size dependence was evaluated on random subsets of $k = 1\text{--}13$ conformations (20 per k), and size bias controlled by repeating the analysis on ligand efficiency.

Pose analysis. Two complementary analyses were performed on the docking results. First, Vina-predicted binding affinities of the five lowest-energy modes were tabulated across all substrates and non-substrates and assessed for their ability to discriminate the two classes when scored either against individual cluster representatives or aggregated across the 13-state ensemble. Second, a geometric and interaction-level analysis was carried out on a focused subset of six experimentally validated CYP3A4 substrates whose primary CYP3A4 sites of metabolism (SOM) are established in the literature: progesterone (STR), erythromycin (ERY), bromoergocryptine (08Y), midazolam (08J), caffeine (CFF), and dronabinol (TCI); their structures and sites of metabolism are shown in Figure 6A. For each of these substrates, the Fe-SOM distance, defined as the Euclidean separation between the heme iron and the SOM heavy atom of the lowest-energy docked pose, was computed across the 13 cluster representatives as a geometric proxy for catalytic competence. Protein-ligand non-bonded contacts, including hydrophobic interactions, hydrogen bonds, and π -stacking, were identified for each docked complex with the Protein-Ligand Interaction Profiler (PLIP) [56] and aggregated per substrate across the 13 representatives to yield interaction-frequency profiles. SOM atoms were assigned from literature-reported CYP3A4 primary oxidation products and verified against the heavy-atom numbering of the corresponding PDBQT file prior to distance calculation. Where the co-crystal structure itself identifies the oxidation site, such as in bromoergocryptine (8'/9' of the proline ring) and midazolam (C1'), the crystallographic assignment was used.

3. Results and Discussion

3.1. Selection of GaMD starting structures

Pairwise C α RMSD analysis across 121 deposited CYP3A4 crystal structures revealed a broad conformational landscape spanning 0.02 to 2.63 Å (Figure 1), reflecting the well-documented plasticity of this enzyme. Three structures were selected to maximally sample this diversity and serve as starting points for GaMD simulations. PDB entry 5TE8 (2.70 Å resolution; midazolam-bound) represents an induced-fit conformation characterized by inward displacement of the F/G loop and a contracted active-site cavity. PDB entry 5VC0 (2.70 Å; ritonavir-bound) defines the opposite extreme: outward F/G loop deflection accompanies a marked enlargement of the substrate access channel, yielding the most open active-site geometry observed in the dataset. The maximum pairwise C α RMSD between these two endpoints reaches 2.63 Å (Figure 1, annotated), underscoring the magnitude of ligand-induced conformational reorganization in CYP3A4. PDB entry 9BV5 (2.71 Å) was selected as an apo-like reference; although a co-crystallized ligand is present, its active-site geometry closely resembles the unliganded state and occupies an intermediate position in the pairwise RMSD map. Together, these three structures span the principal axes of CYP3A4 conformational variability and provide complementary starting geometries from which GaMD can efficiently sample the broader conformational ensemble.

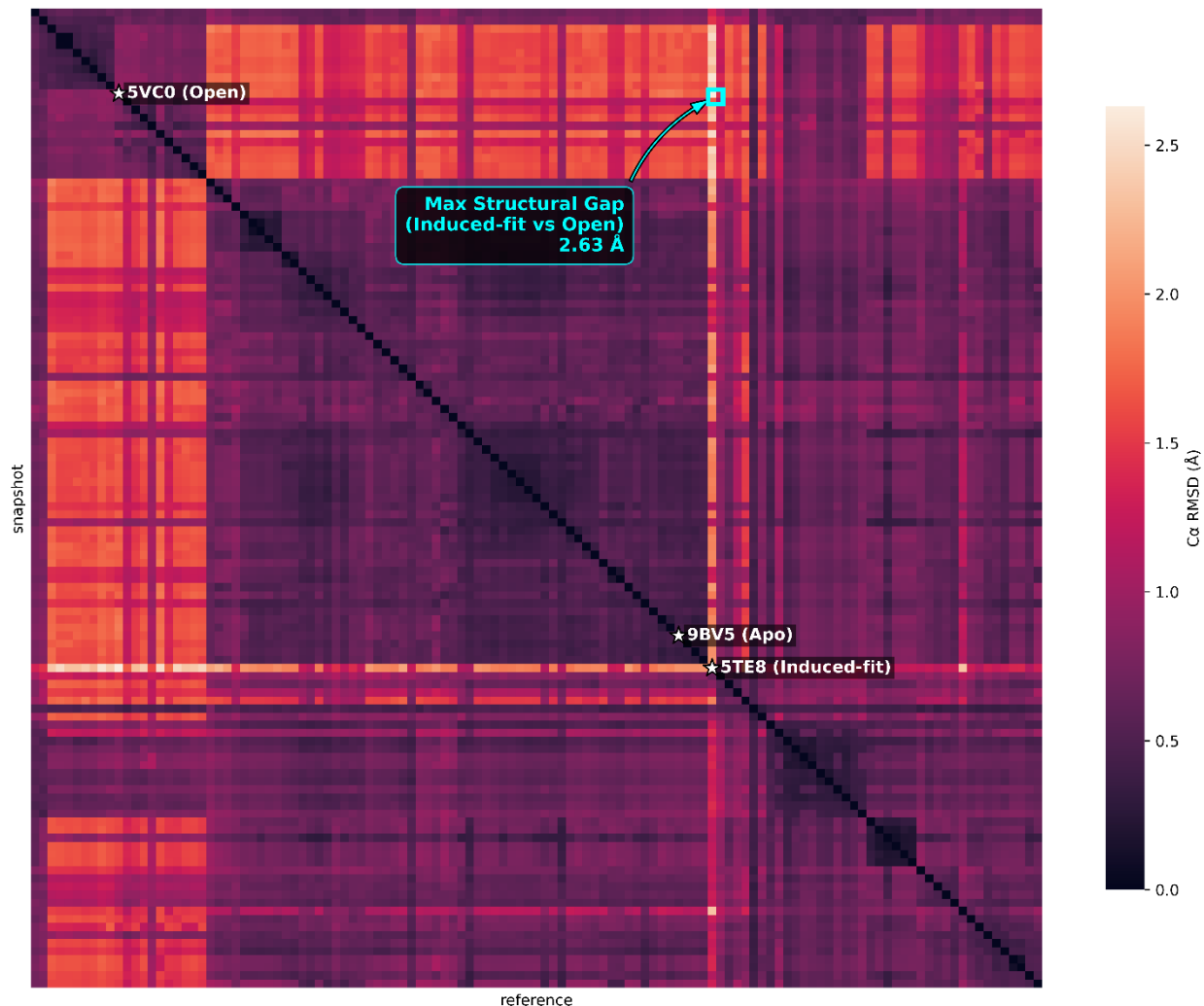


Figure 1. All-to-all pairwise C α RMSD heatmap of 121 experimental CYP3A4 X-ray structures. Color scale: 0 (dark purple) to >2.5 Å (yellow). Distinct conformational families are visible as blocks along the diagonal. The three GaMD starting structures (5TE8, 5VC0, 9BV5) were selected to span the principal axes of this landscape.

3.2. GaMD trajectory analysis

Three independent 500 ns GaMD simulations were performed from each of the three crystallographic starting structures, yielding nine trajectories in total. C α RMSD profiles remained stable throughout all replicates (Figure 2B), with the apo (9BV5) and open (5VC0) systems converging to mean backbone deviations below approximately 2 Å, while the induced-fit system (5TE8) exhibited greater structural drift, reaching values of 3–4 Å by the end of the simulation. This behavior reflects relaxation of the compact, ligand-accommodated geometry upon removal of the bound substrate.

Root mean square fluctuation (RMSF) analysis (Figure 2C) identified the F–F' loop and G–G' loop as the regions of highest flexibility across all three systems, consistent with their established roles in gating the substrate access channel [57], [58], [59]. Several active-site residues, including PHE218, VAL224, LYS256, GLU257, and ILE275, displayed pronounced per-residue fluctuations, pointing to dynamic rearrangements in the vicinity of the heme-binding cavity.

Following initial structural alignment onto the apo-system reference core, principal component (PC) analysis (PC1 = 33.9%, PC2 = 20.0%; Figure 3A) shows that all three systems project from a shared coordinate origin at 0 ns. Each trajectory subsequently converges onto its own distinct region of conformational space: the apo and open systems settle into compact, well-defined clusters by late simulation time, while the induced-fit trajectories converge onto a broader and more diffuse region, consistent with the larger RMSD values observed in panel B. The apo and open ensembles show partial overlap within this two-dimensional projection, but the crossing is confined to a narrow region of PC space and should be interpreted with caution: it may reflect superficial proximity in the reduced representation rather than genuine access to a shared intermediate basin.

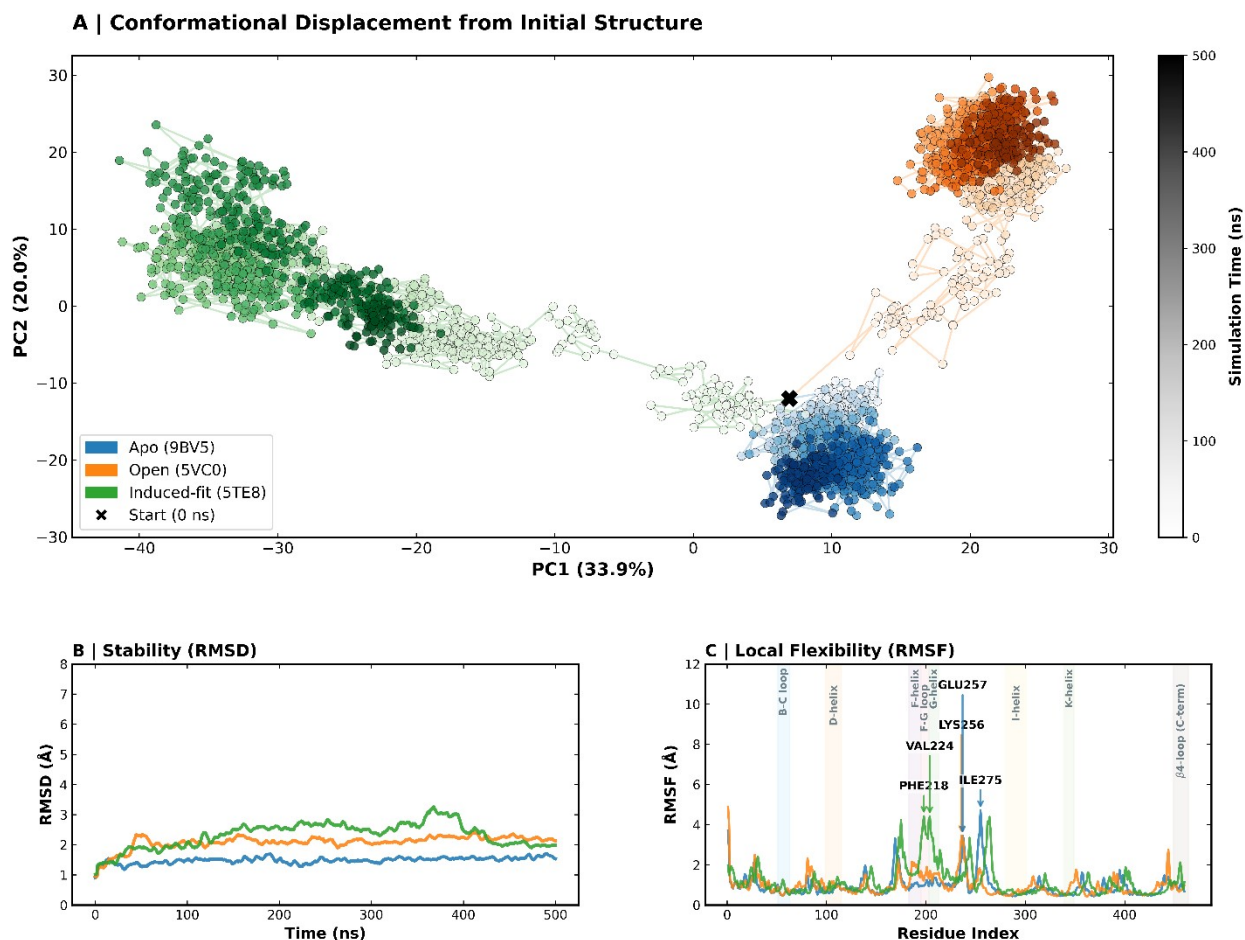


Figure 2. GaMD trajectory analysis of CYP3A4 starting conformations. (A) Backbone PCA (PC1 = 33.9%, PC2 = 20.0%). Frames are colored by system (blue: apo, 9BV5; orange: open, 5VC0; green: induced-fit, 5TE8) and shaded by simulation time (0–500 ns); the black cross marks the 0 ns origin. (B) Time-dependent C α RMSD relative to starting structures. (C) Average per-residue C α RMSF. Shaded regions and labeled residues highlight structurally variable segments and active site flexibility maxima, respectively.

Independent runs showing the dynamic progression of each isolated system are provided in Figure S2. Interpretation of these unliganded trajectories reveals that without steric constraints in the binding pocket, the gating loops undergo spontaneous, large-amplitude breathing motions, as evidenced by distinct localized sub-clusters in Principal Component Analysis (PCA) space and multi-angstrom backbone RMSD shifts across replicates. Conversely, supplementary analysis of

the ligand-bound simulations confirmed the stability of the co-crystallized molecules throughout the induced-fit and open trajectories, with protein and ligand RMSD values remaining below 3.0 Å for the majority of the simulation time (Figure S1). Notably, simulating in the holo state results in significantly less overall conformational exploration compared to the apo systems (Figure S2); the continuous presence of the bound ligands drives active-site compaction and ligand-induced pocket shaping, effectively introducing a barrier that restrains macroscopic loop displacements and locks the enzyme into specific, highly localized functional basins.

3.3. Conformational landscape and ensemble construction

Hierarchical clustering of the 75 merged trajectory snapshots compressed the 4.5 μ s of aggregate GaMD sampling into a tractable 13-state ensemble. Coverage analysis confirmed that all sub-ensembles achieved 100% trajectory coverage at the selected 2.25 Å RMSD cutoff (Figure S3). Specifically, the mean minimum-RMSD to the nearest cluster centroid was remarkably low across the apo (33,600 frames; mean RMSD 1.07 Å, max 1.48 Å; Figure S3A), induced-fit (33,600 frames; mean RMSD 1.17 Å, max 1.92 Å; Figure S3B), and open systems (24,575 frames; mean RMSD 1.05 Å, max 1.61 Å; Figure S3C). Furthermore, the global 13-state ensemble successfully covered all 75 initial representative frames (mean RMSD 1.13 Å, max 1.88 Å; Figure S3D), confirming adequate and comprehensive representation of the sampled conformational space without omitting transient geometric states.

To further characterize this landscape, the 13 cluster representatives were analyzed by binding pocket volume and mapped against experimental data (Figure 3). While derived from three starting structures, the ensemble differentiates into four distinct structural states based on pocket morphology: Closed-Compact (C5–C7; blue), Induced-Extended (C1–C4; orange), Open-Intermediate (C8–C9, C11; green), and Open-Wide (C10, C12–C13; red) (Figure 3A). A structural coverage heatmap (Figure 3B) demonstrates how these states comprehensively map to the 121 deposited X-ray structures. PCA of binding-pocket C α coordinates (75 conserved active-site residues) showed that PC1 (67.6% variance) and PC2 (23.7% variance) together explain 91.3% of total binding-pocket motion (Figure 3C). Projection of experimental CYP3A4 X-ray structures onto this space revealed that every known crystal structure falls well within the GaMD sampling density. The four identified structural states occupy distinct regions of this landscape. Notably, the Open-Wide state (red) extends into regions of PCA space largely unoccupied by any static X-ray structure, representing expanded active-site geometries required for bulky substrate accommodation that were sampled exclusively during simulation.

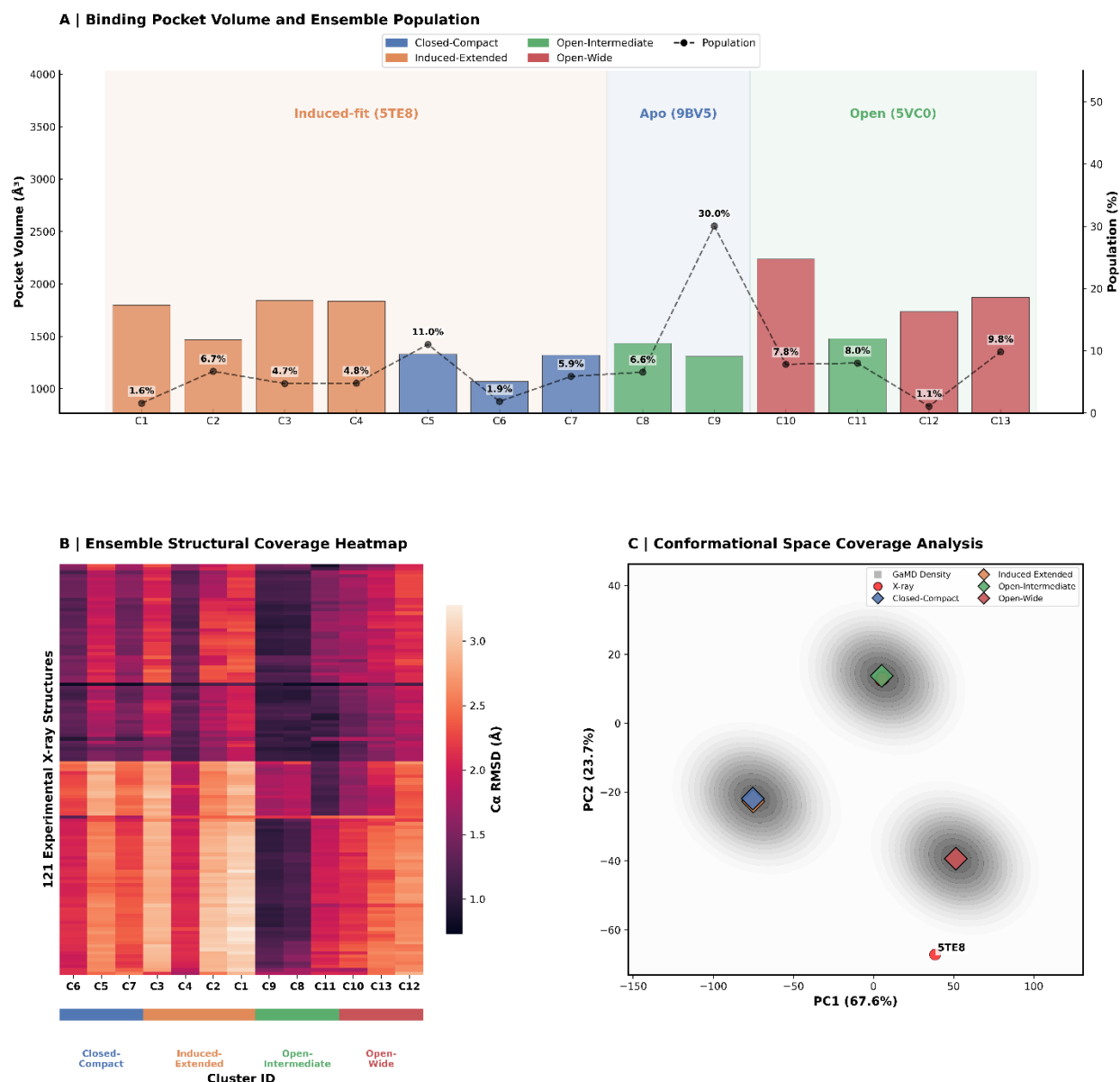


Figure 3. CYP3A4 ensemble structural characterization and conformational landscape. (A) Binding pocket volume (bars) and ensemble population (dashed line) for the 13 cluster representatives. Clusters are grouped by simulation origin (background shading) and colored by defined structural state: Closed-Compact (blue), Induced-Extended (orange), Open-Intermediate (green), and Open-Wide (red). (B) C α RMSD heatmap evaluating structural coverage of the 13 cluster representatives against 121 experimental X-ray structures. (C) Conformational space coverage analysis via PCA of binding-pocket C α coordinates (PC1 = 67.6%, PC2 = 23.7%). GaMD sampling density is shown as grey contours. X-ray structures (red circles) are overlaid alongside the centroids of the four structural states (colored diamonds).

To visualize the structural basis of the four identified states, cluster representatives were superimposed to map the localized domain movements (Figure 4). The most pronounced conformational plasticity occurs in the primary active-site gating elements: the F-F', G-G', and B-C loops. Transitioning from the Closed-Compact state to the Open-Wide state requires a coordinated, outward expansion of these regions, which dramatically lifts the "ceiling" of the substrate access channel. Capturing these massive breathing motions is precisely where robust enhanced sampling protocols excel; accurately resolving these expanded loop geometries is

essential for properly accommodating diverse structural scaffolds, ranging from transient, smaller molecules to rigid, bulky steroids, during downstream docking evaluations.

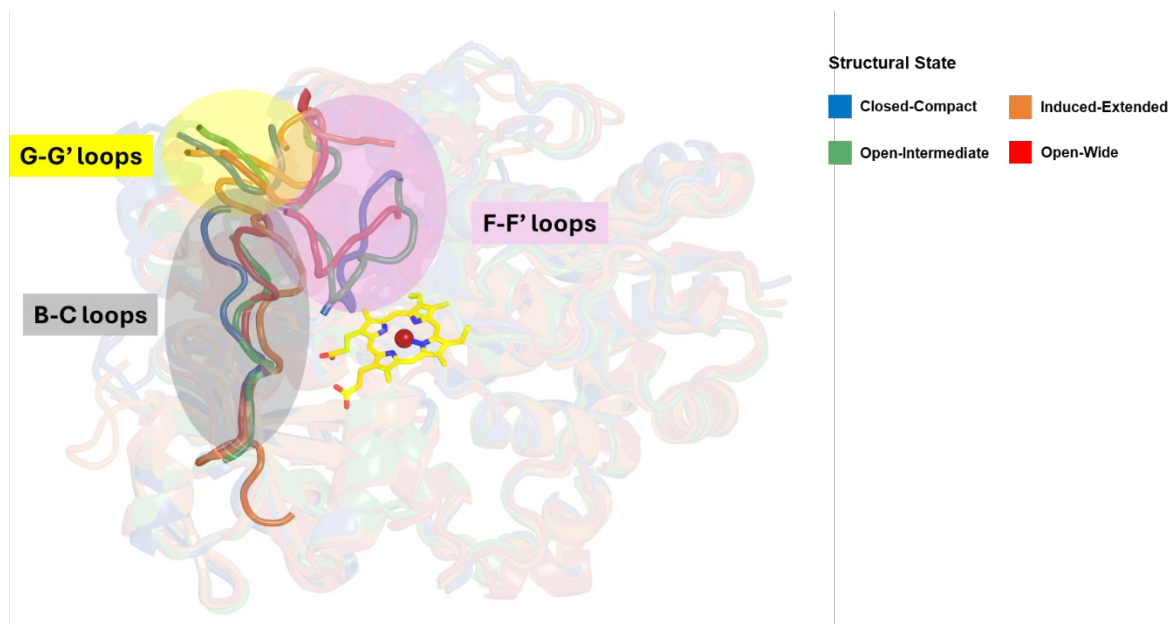


Figure 4. Superposition of CYP3A4 conformational states highlighting active-site gating loops. Representative structures for the Closed-Compact (blue), Induced-Extended (orange), Open-Intermediate (green), and Open-Wide (red) states are overlaid. Shaded ellipses denote the localized outward expansion of the F-F' (pink), G-G' (yellow), and B-C (grey) loops relative to the central heme cofactor (yellow sticks).

3.4. Protein-ligand docking analysis

To systematically evaluate the functional competence of the GaMD-derived structural states to support metabolically productive substrate binding, a curated CYP3A4 library (1,343 substrates and 1,622 non-substrates) was docked across all 13 cluster representatives. The global score-versus-iron-proximity landscape (Figure 5D) reveals a broad distribution of binding geometries across the full compound library, reference substrates including progesterone, erythromycin, and midazolam clustering toward favorable affinities and close heme iron proximity, while caffeine sits apart from the others on both axes, consistent with its low potency.

A logistic regression classifier trained on the 13 per-conformation docking scores separated CYP3A4 substrates from non-substrates with an ROC-AUC of 0.735 (95% CI 0.717–0.754), against 0.697 (0.678–0.718) for the apo conformation C9 used as the single-structure reference (Figure 5A, B; $\Delta\text{AUC} = +0.038$, DeLong $p = 6.7 \times 10^{-8}$).

Individual conformations spanned 0.604 (C6) to 0.731 (C8), with C9, the structure a crystallographic rationale would select, ranking twelfth of thirteen at 0.697 (Figure 5E). Since the best conformation cannot be identified in advance, single-structure docking carries a 0.13 AUC selection risk that the ensemble removes: performance improved sharply at first and then plateaued: 87% of the total gain (0.707 to 0.735) was already captured by $k = 8$, and the spread across random conformation subsets contracted from 0.031 to 0.001 over the same range (Figure 5C). Ensemble docking therefore improves discrimination and, more importantly, makes it independent of structure choice.

Two controls exclude trivial explanations. On a single conformation the classifier reproduces that conformation's raw docking AUC exactly, so the gain is not a product of model capacity. And because Vina scores track molecular size ($r = -0.86$ with heavy-atom count, size alone giving AUC 0.703), the analysis was repeated on ligand efficiency; the ensemble advantage persisted and widened (0.717 vs 0.650, $\Delta\text{AUC} = +0.068$, $p = 1.5 \times 10^{-12}$).

Affinity is a measure of binding, not catalytic competence. Adding the ensemble-averaged iron-to-site-of-metabolism distance improved classification over affinity alone (AUC 0.723 to 0.735, $p = 0.0055$). The discriminating quantity, however, was not the closest approach reached in any single conformation, which carried no information at all (AUC 0.491), but the mean and variability of this distance across the thirteen structures (0.644 and 0.661). Catalytic geometry is thus an ensemble property, inaccessible to any single receptor conformation.

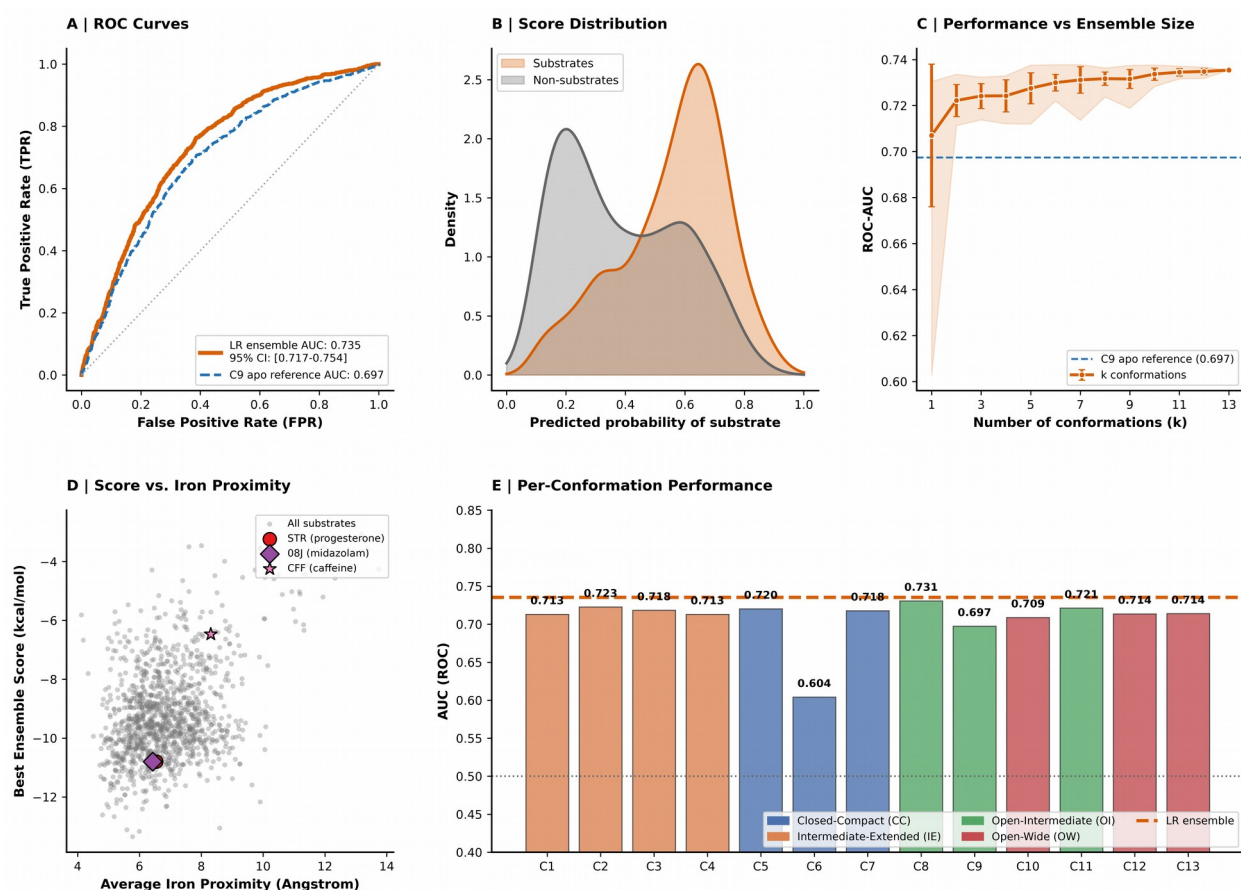


Figure 4. Evaluation of ensemble docking and classification performance. (A) ROC curves comparing the logistic regression ensemble model (solid orange) to the single C9 apo structure (dashed blue). (B) Density plot of predicted substrate probabilities for known substrates (orange) versus non-substrates (grey). (C) Classification performance as a function of the number of conformations included ($k = 1-13$, 20 random subsets per k); shaded band shows the range across subsets. (D) Scatter plot of best ensemble docking score versus average iron proximity, highlighting representative substrates (STR, ERY, O8J, CFF). (E) Per-conformation classification performance (AUC) compared to the ensemble model (dashed orange line).

3.5. Catalytic geometry and productive pose identification

The six reference substrates span a steroid, a macrolide, an ergopeptine, a benzodiazepine, a xanthine, and a cannabinoid, with sites of metabolism ranging from ring hydroxylation to N-demethylation (Figure 6A). Fe-SOM distances for the top-ranked docked pose in each substrate-conformation pair are shown in Figure 6B. Across the six reference substrates, productive poses ($\text{Fe-SOM} \leq 6.0 \text{ \AA}$) were recovered in at least one conformation for every compound, but the identity of the productive conformation varied markedly by substrate, illustrating the complementary coverage provided by the ensemble.

Erythromycin (ERY) presents the most instructive case. The co-crystallized X-ray structure lacks the F-G loop and adopts a non-productive geometry, yet the GaMD ensemble recovers a productive pose with an Fe-SOM distance of $3.1 \text{ \AA}$ in the Open-Wide conformation C10, consistent with productive N-demethylation and directly demonstrating the value of conformational sampling for large, flexible substrates. Bromoergocryptine (O8Y) achieves its optimal catalytic geometry in the Open-Intermediate conformation C11 ($3.6 \text{ \AA}$), reflecting the ability of its bulky ergoline scaffold to exploit the partially expanded access channel; its 8'/9'

proline-ring sites reach a productive distance in eight of the thirteen conformations. Progesterone (STR) is best accommodated in the Intermediate-Extended state C3 (4.9 Å), with a second productive geometry in the adjacent C2 (5.7 Å), consistent with the moderate channel opening required to present the 6 β position of its rigid steroid core to the heme. Midazolam (08J) finds its best productive pose in the Open-Intermediate conformation C11 (5.3 Å), alongside additional productive geometries in neighboring Intermediate-Extended states. Caffeine (CFF), the smallest substrate in the set, achieves the closest Fe–SOM distance of 3.6 Å in the Intermediate-Extended conformation C3, indicating that even compact substrates benefit from partial channel opening to adopt a reactive orientation. Finally, dronabinol (TCI) is optimally accommodated in the Open-Intermediate conformation C9 (3.9 Å), consistent with the preference of its terpenoid scaffold for a moderately expanded binding cavity.

Collectively, no single conformation recovered productive poses for all substrates, and the best catalytic geometry for each compound was distributed across all conformational classes represented in the ensemble. This substrate-dependent partitioning of productive poses provides a structural rationale for the broad and promiscuous substrate scope of CYP3A4 and confirms that explicit conformational heterogeneity is required to faithfully model its metabolic competence *in silico*.

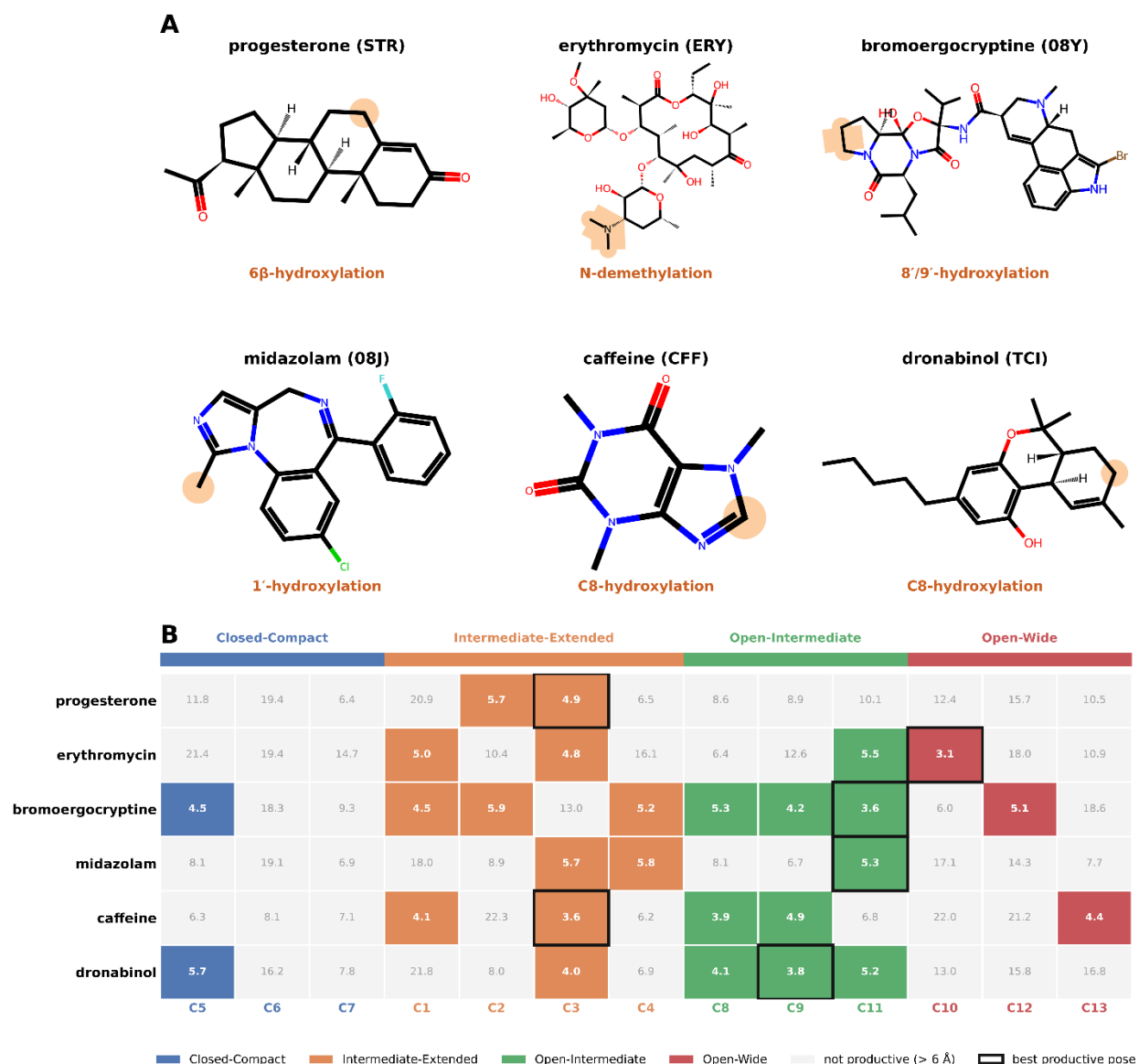


Figure 5. (A) The six reference substrates, with the CYP3A4 site of metabolism highlighted on each: a steroid, a macrolide, an ergopeptide, a benzodiazepine, a xanthine, and a cannabinoid. (B) Fe–SOM distance for the top-ranked pose of each substrate in each of the 13 GaMD conformations, grouped by structural state. Cells are filled in their state color where the pose is catalytically productive (Fe–SOM $\leq$ 6.0 Å) and left pale otherwise; the outlined cell is each substrate's closest approach. Every substrate reaches a productive geometry in at least one conformation, but no single conformation is productive for all six.

To elucidate the mechanisms underlying catalytically competent pose recovery, we performed a comparative structural analysis of erythromycin (ERY) and progesterone (STR), two substrates representing opposite extremes of CYP3A4 pocket accommodation (Figure 7). Across both compounds, GaMD ensemble docking yielded improvements in both predicted binding affinity and catalytic proximity relative to static X-ray docking (Figure 7C–D). Critically, docking against static crystal structures failed to identify productive geometries (Fe–SOM $\leq$ 6.0 Å) for either substrate, whereas the GaMD ensemble placed both well within the catalytic threshold. The structural overlays reveal that these two substrates exploit distinct modes of protein plasticity. For erythromycin, the co-crystallized X-ray reference lacks a resolved F/G loop,

creating an artificial pocket boundary that forces ERY into a solvent-exposed, non-productive orientation (Fe–SOM = 16.7 Å; Figure 7A). GaMD simulations restore the conformational dynamics of this region, providing the structural enclosure necessary to engage PHE108, PHE215, and PHE220 in coordinated aromatic contacts that orient the macrolide to a productive Fe–SOM distance of 3.1 Å. Progesterone illustrates the same principle. Static docking displaces STR far from the heme iron (23.5 Å), whereas an Intermediate-Extended GaMD conformation allows the active-site phenylalanine cluster to rearrange around the rigid steroid core, converging on 4.9 Å at the 6 β position (Figure 7B).

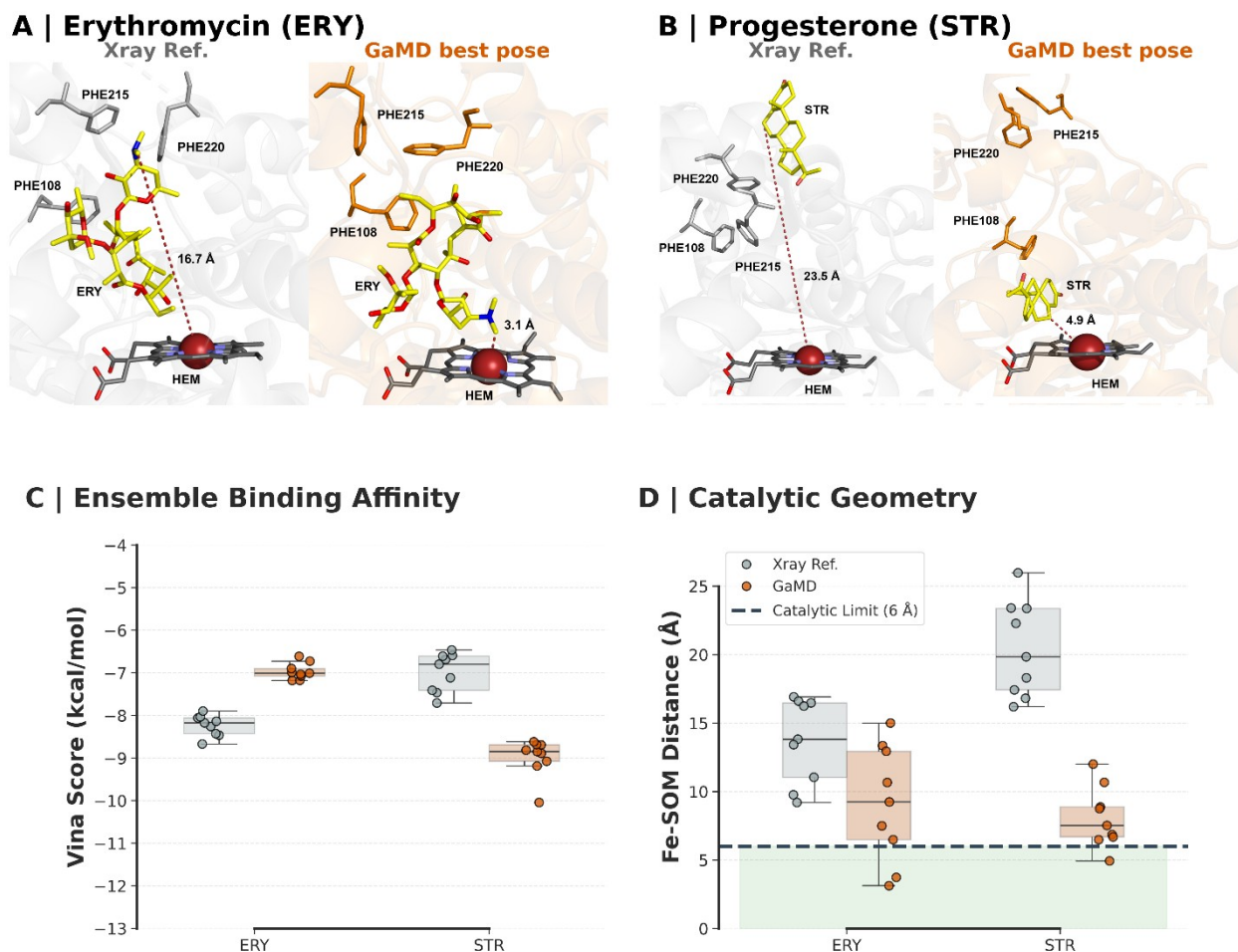


Figure 6. Structural and energetic comparison of erythromycin (ERY) and progesterone (STR) binding in X-ray versus GaMD-derived conformations. (A) ERY in its co-crystal structure (grey, left) and docked into its best GaMD conformation, Open-Wide C10 (orange, right). Highlighted residues (Phe108, Phe215, Phe220) show the role of the restored F–F' loop in orienting the substrate. (B) STR in its co-crystal structure (grey, left) versus the GaMD Intermediate-Extended conformation C3 (orange, right). (C) Box plots of predicted Vina binding affinities (kcal/mol) for the nine docked modes in the crystal receptor (grey) and in each substrate's best GaMD conformation (orange). (D) The same poses scored for catalytic geometry (Fe–SOM distance, Å), measured to the N-demethylation site for ERY and the 6 β position for STR. No pose in either crystal receptor falls below the 6.0Å catalytic limit (dashed line), which both substrates reach in their GaMD conformation.

3.6. Substrate–receptor interaction profiling

Protein–ligand interaction frequencies across the 13-conformation ensemble were analyzed to identify residues responsible for substrate anchoring and catalytic orientation (Figure 8). Three interaction classes, hydrophobic contacts, hydrogen bonds, and π -stacking, reveal a modular and substrate-dependent recognition logic centered on a conserved set of active-site residues.

The phenylalanine cluster spanning the F/G loop and G-helix (PHE108, PHE213, PHE215, PHE219, PHE220, PHE304) constitutes the primary determinant of substrate orientation. Within this cluster, PHE213 functions as the dominant hydrophobic anchor, reaching contact frequencies of 69% for ERY and 62% for 08Y, the two largest substrates in the set. PHE215 and PHE220 provide complementary hydrophobic reinforcement across nearly all compounds, while PHE108 and PHE304 contribute consistent π -stacking interactions (31–38% across most substrates), suggesting conserved roles in trapping aromatic and bulky scaffolds against the active-site floor and ceiling, respectively.

Hydrogen bonding is more substrate selective. SER119 engages ERY and 08Y with frequencies of 31% and 23% respectively, consistent with transient F/G loop contacts during induced-fit accommodation of large macrolide and ergoline scaffolds. ERY additionally recruits ARG212, PHE213, and PHE215 as hydrogen bond partners at appreciable frequencies, reflecting the distributed polar surface of the macrolide ring. In contrast, the smaller and less polar substrates CFF and 08J display minimal hydrogen bond engagement across the ensemble.

Erythromycin exhibits the broadest interaction fingerprint overall, distributing dense hydrophobic and hydrogen bond contacts across the B-C loop, F/G loop, and I-helix simultaneously. This extensive network likely enables the macrolide to remain productively oriented across conformationally diverse active-site geometries. Caffeine displays the sparsest profile, with hydrophobic contacts largely confined to PHE108 and PHE304 and negligible hydrogen bonding, consistent with its characteristically loose binding and high Michaelis constant (K_m). The substrate-specific partitioning of contact frequencies across the ensemble supports a recognition mechanism in which the complete interaction fingerprint assembles dynamically as the active site adapts around the substrate, rather than through rigid pre-formed complementarity.

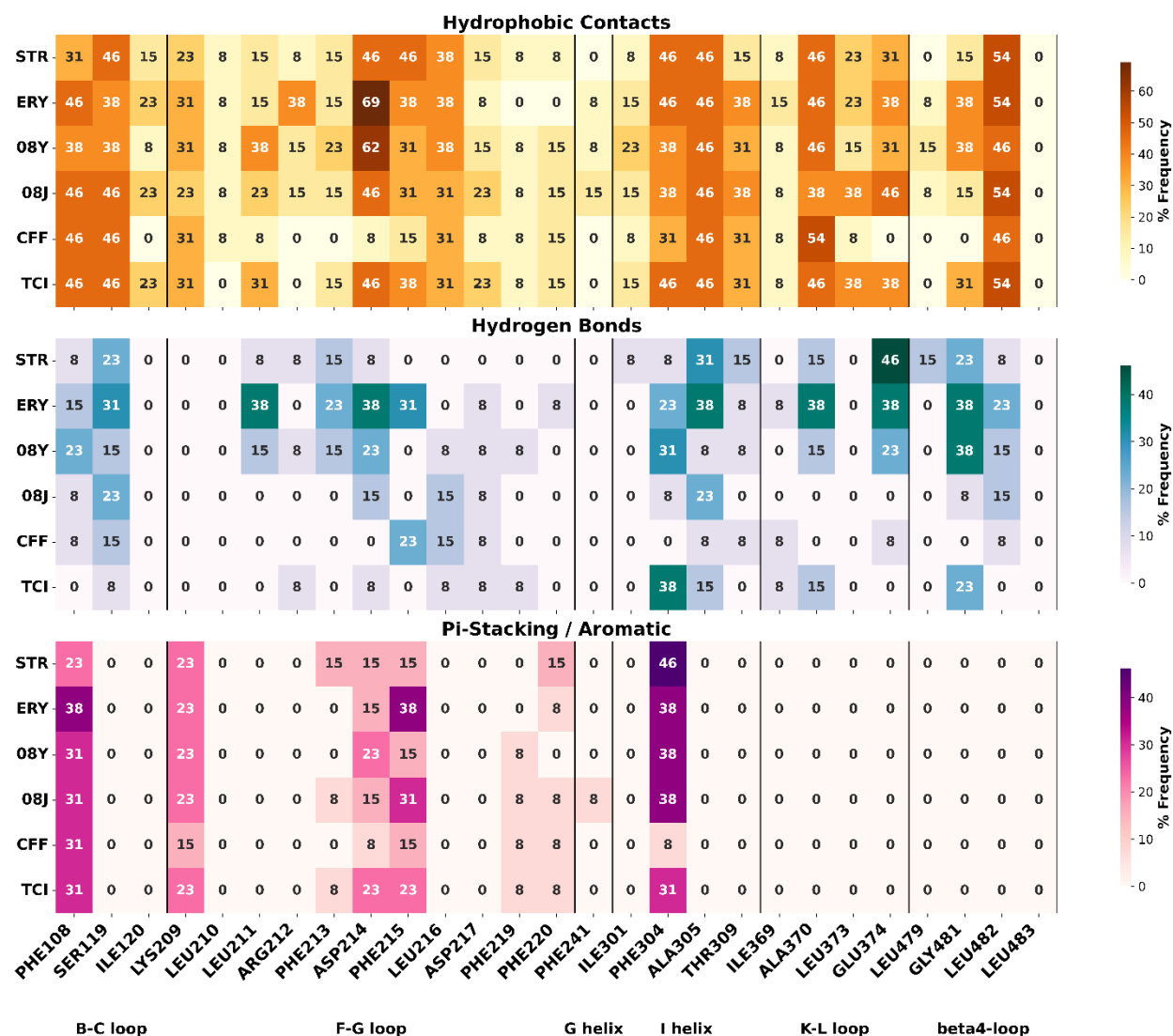


Figure 7. Substrate–receptor interaction frequency heatmaps across the CYP3A4 GaMD ensemble. Rows represent the six evaluated substrates, and columns correspond to key active-site residues grouped by their structural domains (e.g., B-C loop, F-G loop, I-helix). The panels display the percentage of the 13 conformations in which each substrate exhibits hydrophobic contacts (top, orange), hydrogen bonds (middle, teal), and π -stacking/aromatic interactions (bottom, pink). Darker shades indicate higher interaction frequencies.

4. Conclusion

This study demonstrates that GaMD-derived conformational ensembles provide a robust foundation for *in silico* prediction of CYP3A4 substrate metabolism. By sampling the enzyme's conformational landscape from three distinct crystallographic starting points, the simulations captured the full range of experimentally observed conformations and revealed additional, transiently populated active site geometries. The most notable of these transient geometries is the Open-Wide state that gives bulky macrolides such as erythromycin their closest approach to the heme, a geometry inaccessible to static crystal structures. Ensemble docking across the resulting 13-conformation repertoire recovered catalytically competent poses for all evaluated substrates, demonstrating that catalytic geometry is a property of the ensemble, rather than any single

receptor conformation. Substrate-specific interaction fingerprints also provided mechanistic insight into how CYP3A4 leverages a modular phenylalanine-centered recognition network to achieve broad binding competence across chemically unrelated scaffolds. At the classification level, a model trained on ensemble docking scores achieved an ROC-AUC of 0.735, outperforming every individual conformation and removing the dependence of performance on which single structure is chosen. This advantage is not an artefact of molecular size: repeated in ligand-efficiency space, where each score is normalized by heavy-atom count, the ensemble reaches 0.717 against 0.650 for the apo reference, a wider margin than in affinity space.

Data and Software Availability

Test compounds were taken from the curated CYP3A4 reference set of Ni et al. [52]; SMILES with class labels are provided as a machine-readable .csv in the Supporting Information. CYP3A4 crystal structures were retrieved from the RCSB Protein Data Bank (<https://www.rcsb.org>) under UniProt accession P08684.

The 13 GaMD cluster representatives, docked poses, per-conformation docking affinities, Fe–SOM distances, and the analysis and plotting scripts are deposited at Zenodo (DOI: [10.5281/zenodo.22794614](https://doi.org/10.5281/zenodo.22794614)) and are sufficient to reproduce all results reported here. Full GaMD trajectories are available from the corresponding author on request owing to their size.

All software used is open source: GaMD-OpenMM (<https://github.com/MiaoLab20/gamd-openmm>), AmberTools for MCPB.py, tleap and cpptraj (<https://ambermd.org>), PyMOL, Meeko v0.7.1, AutoDock Vina v1.2.6, PLIP, and scikit-learn v1.9.0. No proprietary or commercial software was used.

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Conflicts of Interest

The authors declare no conflicts of interest.

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